Xi'an Jiaotong-Liverpool University

Xi'an Jiaotong-Liverpool University 西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT 104		Department of Computing	

2nd SEMESTER *2024/25 RESIT* EXAMINATION

OPERATING SYSTEMS CONCEPTS

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This is an open-book examination.
2. Total marks available are 100, accounting for 100% of the overall module marks.
3. Answer all FOUR questions.
4. The number in the column on the right indicates the marks for each question.
5. Relevant and clear steps should be included in the answers.
6. The university approved calculator - Casio FS82ES/83ES can be used.
7. Only English solutions are accepted.
8. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

QUESTION I. Fundamentals (38 marks)

1. Why are programs and data not resided in main memory permanently? Give **two reasons** and explain your answer.

(6 marks)

2. What are **two advantages** of encrypting data stored in the computer system? Explain your answer

(6 marks)

3. What is the function of the **ready queue**? (3 marks)

4. Describe the general strategy behind **deadlock prevention** and give an example of a practical deadlock prevention method.

(6 marks)

5. Compare I/O based on **polling** with **interrupt-driven I/O**. In what situation would you favour one technique over the other? Explain your answer.

(8 marks)

6. Let us say you are a database administrator for a company. Your job is to store employee records in a file that allows the records to be accessed and updated randomly. Each employee's record is of fixed size. There are three methods of allocating disk space: *contiguous*, *linked* and *indexed*. Which of these will be most appropriate? Explain your answer.

(9 marks)

QUESTION II. CPU scheduling, Memory management, Disk scheduling

(38 marks)

1. Consider the following scenario of processes. The processes are assumed to have arrived in the order A, B, C, D, E, all at time 0. Their burst time and priority are as follows:

Process	Burst time (ms)	Priority
A	10	3
B	6	1
C	2	4
D	4	5
E	8	2

a) Draw the **Gantt chart** for the execution of the processes and calculate **the average turnaround time** for the system using the **Round Robin scheduling algorithm** (time quantum = 2ms).

(7 marks)

b) Draw the **Gantt chart** for the execution of the processes and calculate **the average turnaround time** for the system using the **Priority scheduling algorithm**.

(7 marks)

2. Calculate the number of page faults for the following sequence of page references (each element in the sequence represents a page number) using the **First-In, First-Out (FIFO)** replacement algorithm with frame size of 4.

0, 1, 7, 2, 3, 2, 7, 1, 0, 3

(6 marks)

3. Consider a disk queue with I/O requests on the following cylinders in their arriving order:

87, 160, 40, 140, 36, 72, 66, 15

We assume a disk with tracks numbered 0 to 180 and the head is initially at position **60**.

Write the sequence in which the requested tracks are serviced using the **Shortest Seek Time First (SSTF) algorithm** and calculate the **total head movement** (in number of cylinders) incurred while servicing these requests.

(6 marks)

4. Consider a **single level paging scheme**. The virtual address space is 16 GB and the page table entry size is 4 bytes. What is the **minimum page size** possible such that the entire page table fits well in one page?

(12 marks)

QUESTION III. Resource allocation (12 marks)

Consider the following snapshot of a system:

Available

A	B	C	D
6	4	4	2

Process	Max				Allocation			
	A	B	C	D	A	B	C	D
P0	3	2	1	1	2	0	1	1
P1	1	2	0	2	1	1	0	0
P2	3	2	1	0	1	0	1	0
P3	2	1	0	1	0	1	0	1

a) What is the content of the **Need** matrix denoting the number of resources needed by each process? (2 marks)

b) Is this system in a **safe state**? If your answer is *yes*, please give a safe sequence and resources available after each process finished. If your answer is *no*, please specify the processes that might involve in a deadlock (unsafe). (5 marks)

c) If a request from **P3** arrives for **(2, 1, 0, 0)**, can that request be safely granted immediately? Explain your answer.

(5 marks)

QUESTION IV. Operating System in C Language (12 marks)

The following C code implements a shared counter accessed by multiple threads. Two semaphores (*sem1* and *sem2*) are used for synchronization.

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

sem_t sem1, sem2;
int counter = 0;

void *thread_A(void *arg) {
   sem_wait(&sem1);
   counter += 5;
   printf("Thread A: counter = %d\n", counter);
   sem_post(&sem2);
   pthread_exit(NULL);
}

void *thread_B(void *arg) {
   sem_wait(&sem2);
   counter *= 2;
   printf("Thread B: counter = %d\n", counter);
   sem_post(&sem1);
   pthread_exit(NULL);
}

int main() {
   pthread_t t1, t2;
   sem_init(&sem1, 0, 1);
   sem_init(&sem2, 0, 0);

   pthread_create(&t1, NULL, thread_A, NULL);
   pthread_create(&t2, NULL, thread_B, NULL);

   pthread_join(t1, NULL);
   pthread_join(t2, NULL);

   sem_destroy(&sem1);
   sem_destroy(&sem2);
   return 0;
}
```

a) Explain the purpose of *sem1* and *sem2*. What is the final value of *counter* after execution?

(4 marks)

b) If the initial value of *sem1* is set to 0, what issue occurs? Explain why. (4 marks)

c) Propose an alternative synchronization mechanism to achieve the same result without semaphores. (4 marks)

END OF EXAM PAPER